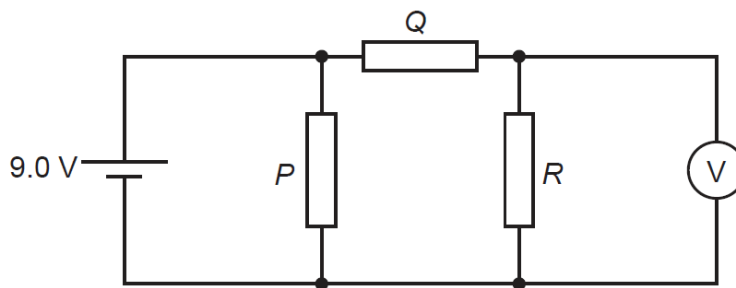


23

A 9.0 V battery with negligible internal resistance is connected into a circuit, as shown.



The resistance of the voltmeter is equal to the resistance of R .

The reading on the voltmeter is 3.0 V.

What is the value of the ratio $\frac{\text{resistance of the voltmeter}}{\text{resistance of } Q}$?

A

 $\frac{1}{2}$

B

1

C

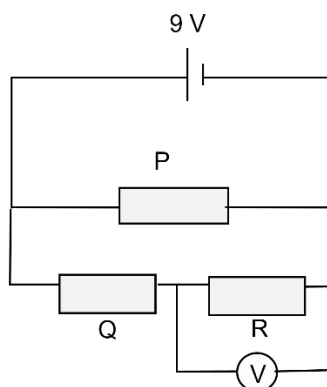
2

D

4

Answer: B

Redrawing the circuit,



Resistor Q and R are in series and have a pd of 9 V across them.

Voltmeter and resistor R are in parallel, the potential difference across both is also 3 V.

Potential difference across Resistor Q is $9 - 3 = 6$ V

Resistance of voltmeter cannot be ignored.

Let resistance of voltmeter be R

Effective resistance of voltmeter and resistor R (as they are in parallel and have the same resistance) = $\frac{R}{2}$

By potential divider approach,

$$\frac{\text{effective resistance of the voltmeter \& resistor } R}{\text{resistor } Q} = \frac{\text{pd across voltmeter \& resistor } R}{\text{pd across resistor } Q}$$

$$\frac{\frac{R}{2}}{Q} = \frac{3}{6}$$

$$\frac{R}{Q} = 1$$

[Turn over

	* If students pick A, they have ignored the resistance of the voltmeter.
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24	Two parallel, straight conductors, placed at right angles to a ruler, carry currents of $2I$ and I .
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